

Solara: application specification

Screening of sites and areas for solar and wind generation, on public data

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Abstract

This document specifies Solara, a desktop application for screening sites and areas for solar and wind generation. At a site, Solara estimates the solar resource, the photovoltaic yield and the wind at hub height. Over an area, it estimates the irradiation on the terrain, where a plant would connect to the transmission grid, the consumption and the fraction of the ground that could hold a plant. The document defines the scope, the requirements, how a reading is obtained, the six products, the screen, the architecture, the data and the caveats that accompany the numbers.

Version	Date	Change
0.2	2026-10-05	First specification of Solara. Replaces version 0.1, which specified a different product, for distribution utilities.

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1 Scope

Solara is a screening tool for the study of given sites and areas, built on public data and under stated assumptions. A *site* is a point on the map; an *area* is a published administrative boundary. At a site, Solara reads the solar resource, the photovoltaic yield and the wind at hub height. Over an area, it reads the plane-of-array irradiation on the terrain, the point of connection to the transmission grid, the consumption and the fraction of the ground that could hold a plant. Over the whole map, it shows the weather at the time of the query: clouds, rain and wind.

Three properties delimit what belongs to the application:

1. **It runs on the user's machine.** It is a desktop application, with no server and no remote account. The computation runs in a local Python process, the *sidecar*, and the Brazilian electricity registry is read from a local PostGIS database, the *Grid store*, hereafter the *store*.
2. **It reads public data.** The sources are open, fetched on demand and cached. No product depends on customer data.
3. **Every result states where it came from.** The result is saved with the parameters and the place for which it was computed, and each number carries the qualification that bounds its interpretation.

The following are out of scope: telemetry and continuous metering, billing and market settlement, demand response, and any function that requires an always-on server, state shared between users or private data. No product computes a forecast of its own; the forecast shown on screen is a third-party one (section 4.2) and is marked as modeled.

2 Requirements

- R1 The map is the surface on which the place is chosen and on which an area result is drawn. An area product that generates a raster draws its layer with a legend, and at the end of the run the map is moved to the place of the reading if that place is not in view.
- R2 The place of a reading is a site or a catalog area. Areas are not drawn freehand (section 3.2).
- R3 A menu lists only what has an answer with the connected store and over the map extent in view. The footprint layer of each registry, on by default, shows where consumption data exist. A product that cannot run states the reason before the run, rather than failing after it.
- R4 The card of a product lists everything the run reads, with the current value and the state of each input. The run reads all the listed inputs, wherever it is started from.
- R5 Every result stays in the project with the parameters and the place for which it was computed. A later change to the place or to a parameter marks the result as stale; the result is not relabeled.
- R6 A modeled value is not drawn as a measured one. The qualification that changes how a number is read (raw, not validated, at least, injected and not generated, not confirmed) sits next to the number.
- R7 Quantities of different kinds are reported separately. Grid proximity and curtailment are not combined into a score.
- R8 Solara reads the store and does not write to it. Without the store, the products that do not depend on it work, and those that do report that it is unreachable and why.
- R9 Consumer units are counted within an area and are not drawn at an address. The coordinate in the registry is that of the point of connection to the distribution grid, shared by the units on the same pole or transformer.
- R10 Every action of the interface is an interface operator: reachable by menu, key, search and console, with a precondition that states why it cannot run. Every change is an undo step.

3 Interaction model

3.1 Parts of a reading

A *product* is a question that Solara answers; a *run* is one computation of the product for a place; a *reading* is the presentation of the result. Every run answers three questions, by user choice or by default value.

Table 1

The three parts of a run.

Part	Question	How the user provides it
Where	for which place?	marks a site on the map or chooses an area in the catalog; the active place is the one of the next run
What	which product?	opens the product from the command strip, from the card in Properties or from the empty reading of the workspace
With what	which parameters and which registry?	fills in the fields of the card; an empty field takes the default of the computation; the store is the connected one

The scale of the place defines which products apply.

Table 2

Scale of the place and what is read at each one.

Scale	What it is	What is read
Site	a point, with longitude and latitude	solar resource, wind screening, current conditions
Area	a catalog boundary, from the municipality upwards	solar terrain, grid connection, area consumption, usable ground, current conditions
Map	the extent in view	store layers, published registries, clouds, rain and wind field

3.2 Area by catalog, not by drawing

The area is not drawn by hand. A drawn polygon can extend beyond the boundary of the registry that answers the question, by an extent that no step checks. For an area drawn in central Ceará, the registry covered 7.4 % of the ground asked about and returned 73,993 consumer units; the reading reported only which registry had been read. An administrative boundary, in contrast, is published, has the same shape for everyone and is the shape over which official figures are computed, so the reading can be set next to a published figure.

Table 3

Area catalogs and the order of preference among them.

Catalog	Covers	Limit of the source
IBGE	states and municipalities of Brazil, in the highest-resolution mesh	the municipality is the floor: the service does not subdivide it, so a neighborhood is not an area
Store	the levels that the store declares in <code>solara.boundary</code>	whatever the person who prepared the store loaded
geoBoundaries	subdivisions of any country, simplified files	the outline misses the border by a few hundred meters; used where the other two do not exist

3.3 Sequence of a run

The run is started from the product card in Properties or from the empty reading, which shows the same list of inputs. The run graph draws the same inputs as nodes and wires, computed by the same functions, but the run is not started from it. A wire that enters the run node cannot be cut, and no cut changes a run. The two wires of the map band, from the region and from the layer to the map, can be cut: with the cut, the layer is drawn over the whole footprint of the registry, without the clipping to the region.

Table 4

State of an input in the product card and in the run graph.

State	Meaning
not set	the input has no value: no active place, store not connected
pending	the value changed after the most recent run
reading	a run is in progress
read	the most recent run read this value
error	the last attempt failed at this input

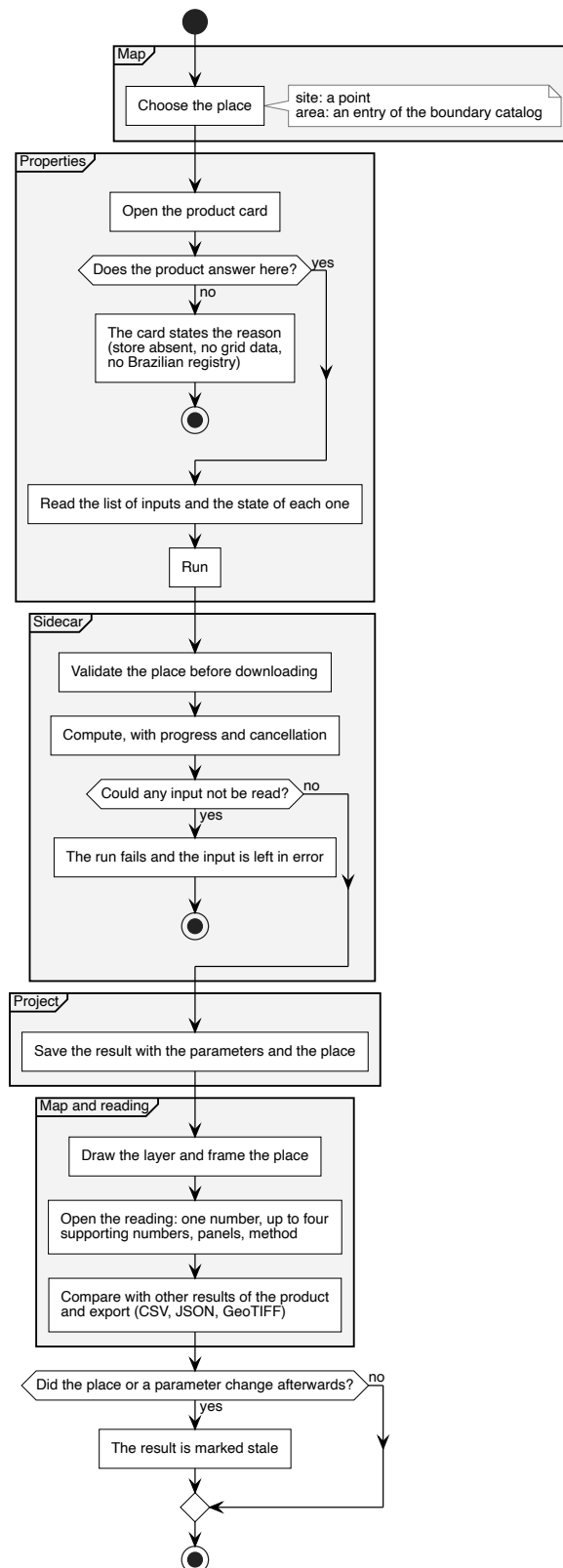


Fig. 1. The run starts from the place chosen on the map and returns to it as a layer. Sequence reconstructed from the interface and the sidecar; each gray band is the part of the application in which the step occurs. The two side exits end without a result: the product does not answer there, or an input could not be read.

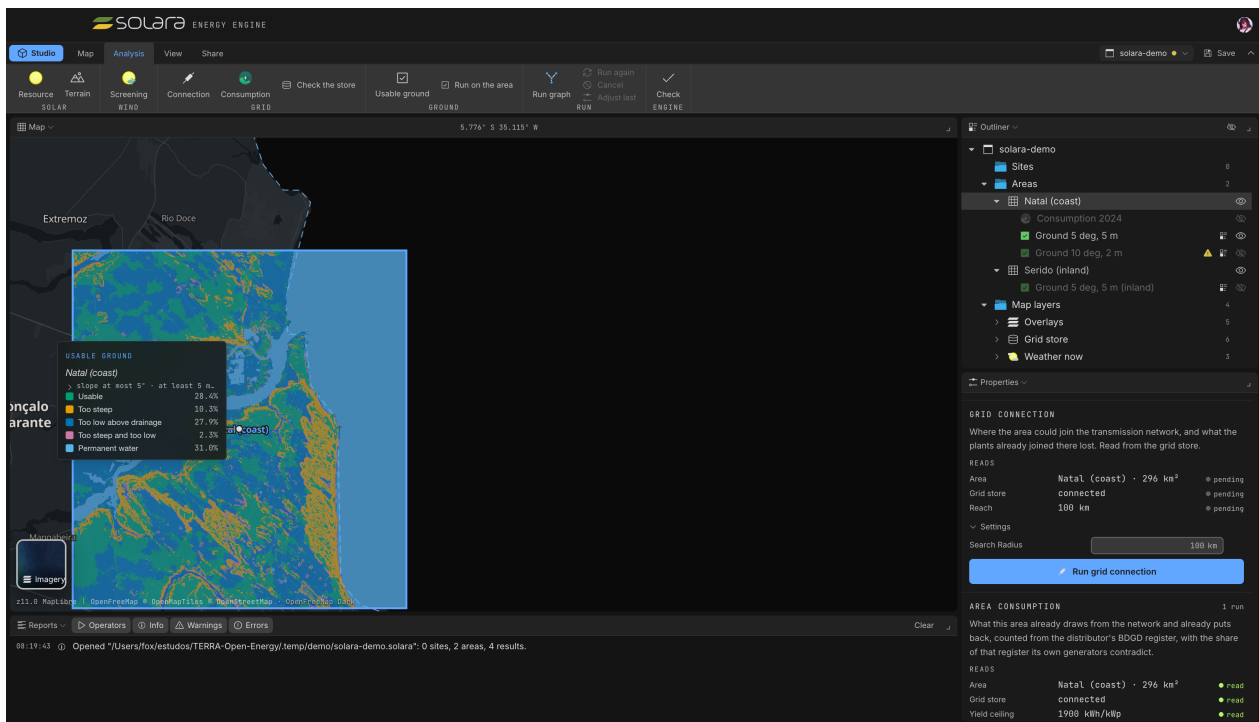


Fig. 2. The product card lists what the run reads and the state of each input before the run. Properties with the area selected: the grid connection card, not yet run, with its three inputs as “pending”, the search radius parameter and the run button; below it, the area consumption card, already run, with its inputs as “read”. Screenshot of the demonstration project, like the others in this document; the area “Natal (coast)” is a four-vertex rectangle saved in the project file, and not a catalog entry.

3.4 Layout of a reading

All readings have the same layout, in this order:

1. the product and the status of the result, the number with which the product answers, and a line that states where and from what;
2. up to four supporting numbers, each next to what it is read against;
3. the panel that breaks down the answer, at full width;
4. the other panels, two per row, or one when the screen area is narrow;
5. how the reading was obtained.

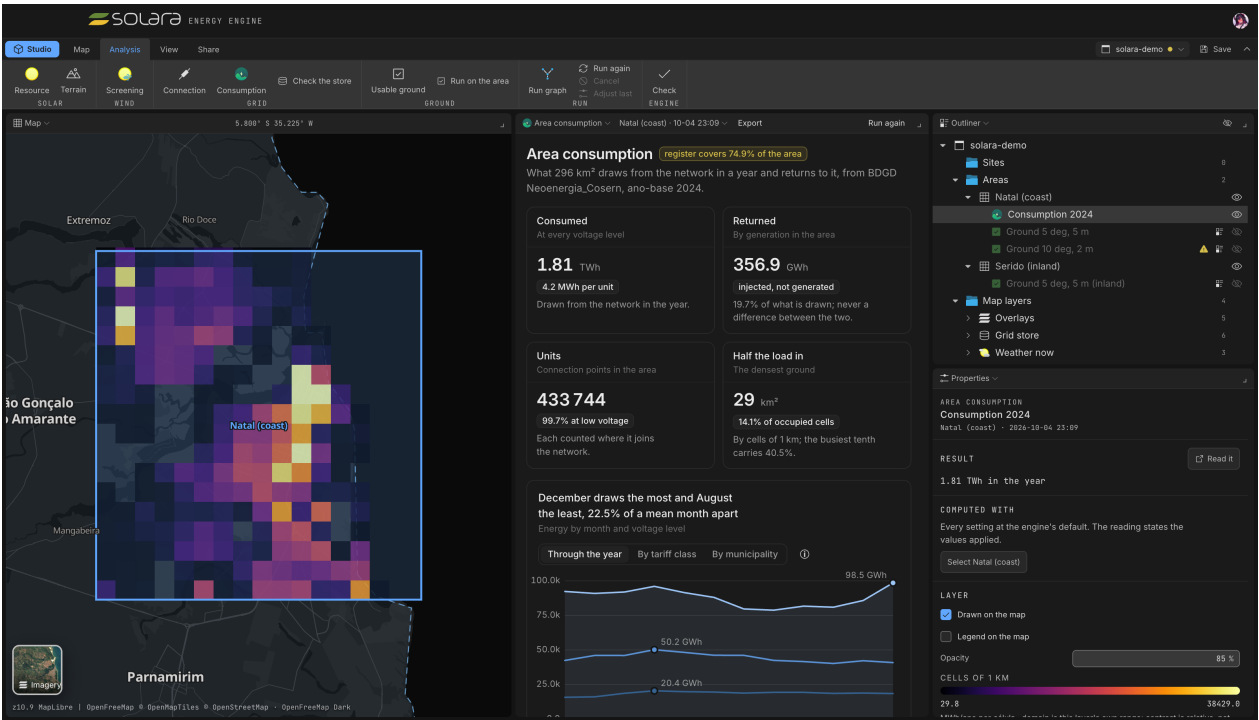


Fig. 3. The reading opens with one number, four supporting numbers and the panel that breaks down the answer. Area consumption over Natal, BDGD of Neoenergia Cosern, base year 2024: the coverage of the registry next to the title, the injected energy with the qualification “injected, not generated” next to the number, and the panel of energy by month and voltage level. On the map, the 1 km consumption grid. Screenshot of the same demonstration project.

The explanation of a panel is opened by the information button in the header of the panel. Properties summarizes a result in one line, with the number of the reading and the button that opens it. Results of the same product are compared side by side in a table and exported as CSV or JSON. The layer of a result is also exported as GeoTIFF: the solar terrain layer in float32, the usable ground layer with the classes. The time of the computation appears in the reader’s time zone on screen, in full UTC in the JSON and in local time with the offset in the CSV.

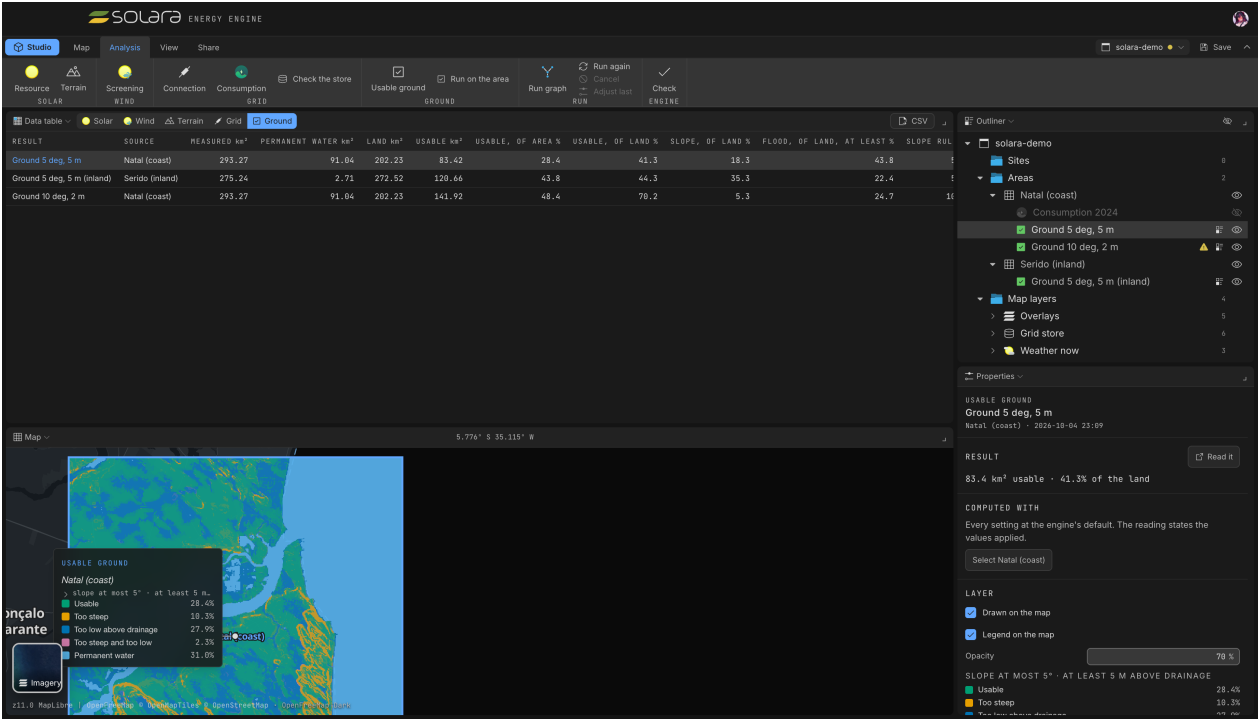


Fig. 4. Results of the same product are compared row by row. Data table with three usable ground results: two rules over the same coastal area and the default rule over an inland area. Screenshot of the same demonstration project.

4 Products

Table 5

The six products, under the names used in the interface.

Product	Read at	What it answers	Source
Solar resource	site	global horizontal irradiation by month and by year, P10 and P90, optimum tilt, specific yield and capacity factor	NASA POWER, pvlib
Wind screening	site	wind at hub height, how much it varies with the assumed shear, Weibull distribution, density, power curve and wind rose. Raw and not validated	NASA POWER (MERRA-2)
Solar terrain	area	plane-of-array irradiation on the terrain, at 30 m, with horizon shading, by season	Copernicus DEM GLO-30, NASA POWER
Grid connection	area	the bus to which the plants of the area are connected, the headroom at that bus, what is within reach and how much the plants already connected there lost to curtailment	ONS and ANEEL registries, in the store
Area consumption	area	what the area draws from the grid over the year, by voltage level, tariff class and municipality, what distributed generation injects, and the distribution on a 1 km grid	BDGD of the distribution utility, in the store
Usable ground	area	how much of the area can hold a plant and what excludes the rest, by maximum slope and minimum height above the nearest drainage (HAND). Permanent water is removed before the two rules	Copernicus DEM GLO-30, ESA WorldCover 2021

4.1 Conditions of each product

- **Solar resource, wind screening and solar terrain** read global sources and are not blocked by the store.
- **Solar terrain** validates the area before downloading any data and limits it to four million cells, about 55 km × 55 km.
- **Grid connection** requires a store with lines and substations. The reading opens with the bus to which the plants of the area are connected, as ONS publishes it, and not with the nearest substation: ONS publishes the 500, 230 and 138 kV buses of a substation at the same coordinate, and sorting by distance selects the wrong voltage.
- **Area consumption** requires the Brazilian registry in the store and refers to one distribution utility at a time. Before the run, the card reports how much of the area the registry covers.
- **Usable ground** applies two rules typed by the user: maximum slope, 5° by default, and minimum height above drainage, 5 m by default, where zero turns the rule off. Without the water map the run fails. The usable fraction is given twice, of the area and of the land within it. Land cover, protected areas and distance to the connection are not read.

4.2 Weather at the time of the query

Solara offers four weather readings, each marked as *observed* or *modeled* and shown with its age. The controls sit on a timeline below the map.

Table 6

Weather readings. All of them depend on the internet; without it, each one reports the failure and the rest of the application is not affected.

Reading	Kind	Source and cadence
Clouds	observed	GOES-East through NASA GIBS, every 10 min, true color or infrared, published with a delay of about 40 min; only within the disk of the satellite
Rain	observed	RainViewer radar composite, every 10 min; where there is no radar, absence of color is not absence of rain
Wind	modeled	GFS at 10 or 100 m, over the map extent in view, as a color field with particles and a label at each place of the project
Current conditions	modeled	Open-Meteo at a site or area: irradiance, cloud, temperature, wind at 10, 80 and 120 m, the next 24 h, and the comparison with the long-term readings of the project

Clouds and rain can be replayed over the last two hours.

5 Screen

The window is a studio: the screen is divided into screen areas, each with an editor, and the arrangements are grouped by subject into workspaces. Screen areas can be split, joined, resized and maximized, and the arrangement is kept across restarts.

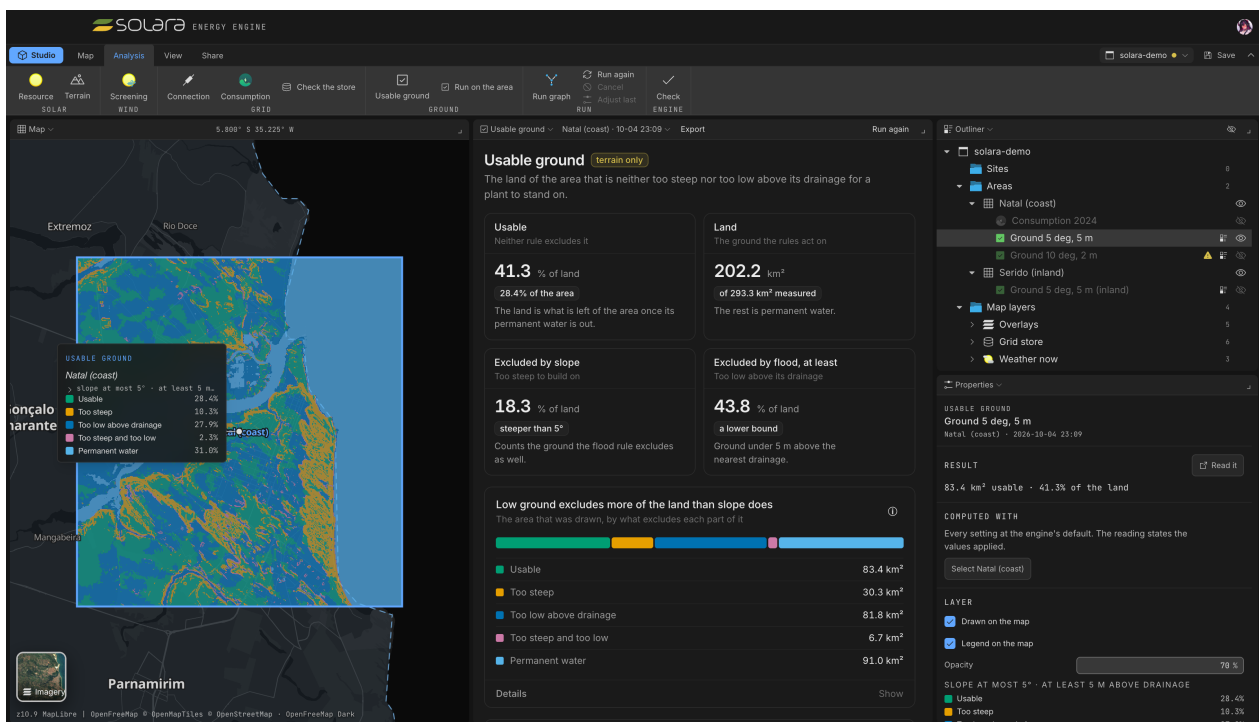


Fig. 5. The map, the reading and the properties of a result sit in the same window. Usable ground over Natal: the class layer on the map with the legend attached to it, the reading in the center with the usable fraction of the land and of the area, and on the right the outliner and Properties with the result in one line. The yellow triangle in the outliner marks a stale result. Screenshot of the same demonstration project.

Table 7
Workspace groups.

Group	Workspaces	Organized around
Board	Layout, Graph, Data, Scripting	map, run graph, data table, console
Solar	Resource, Terrain	solar readings next to the map
Wind	Screening	wind reading next to the map
Grid	Connection, Demand	connection and consumption readings next to the map of plants and lines
Ground	Usable ground	usable ground reading next to the map, with the class layer

The editors are the map, the outliner, Properties, the run graph, one reading per product, the data table, the reports and the console.

5.1 Map layers

Table 8
Map layers and the condition under which each one is offered.

Layer	Content	Offered when
Plants in the operational registry	size by capacity	store with the Brazilian registry
Plants only registered	those that the operational registry does not cover	store with plants
Lines and buses	lines colored by voltage, in the ANEEL convention	store with grid data
Footprint of each registry	union of the BDGD sets (<i>conjuntos</i>) of each distribution utility; on by default	store with the Brazilian registry
Consumption by municipality	all the municipalities of each registry	store with the Brazilian registry
SIGEL	wind turbines one by one and strips declared of public utility	map over Brazil
Night lights	NASA Black Marble, as a check of where there is load	always
Basemap	dark street map or cloudless Sentinel-2 imagery of 2016, with shaded relief under the other layers	always
Area result	raster of the solar terrain or of the usable ground classes, with legend, visibility and opacity	there is a result in the project

Clicking a feature of a store layer opens a legend at the clicked point. The SIGEL and night lights layers arrive as images from the services themselves: they report where something is, not what it is, and cannot be clicked. The credit in the footer of the map follows the basemap in use and the registries that are on.

6 Architecture

Table 9

Components and the responsibility of each one.

Component	Responsibility
Interface (webview)	the studio: project state, map, editors, interface operators, undo
Go/Wails shell	the calls that the interface makes; what each product sends and the shape of what comes back; the project file; the results directory of the session, served to the interface
Python sidecar	the computation. One interpreter per request: the request enters as JSON on standard input, the progress leaves on standard error, the result leaves as JSON on standard output. One request at a time, cancellable
Grid store	PostgreSQL with PostGIS on the same machine, read by the sidecar
Project	a file <code>name.solara</code> , with the terrain rasters in a folder <code>name.solara-data</code> next to it

The sidecar protocol is that of TERRA, the program that loads the `br` schema of the store, and the actions have the names and the result shapes of TERRA where the two programs share a product, so that an action moves from one to the other without change. The raster of a result is served from the results directory and does not travel as a data URI.

Table 10

Where each item is stored.

What	Where
local accounts and the chosen store	user configuration directory
NASA POWER series and the wind field of the hour	user cache directory
terrain rasters of the session	temporary directory, removed on exit; saved with the project
studio arrangement, layers that are on, recent files	local storage of the webview

7 Data

7.1 Grid store

The store is a PostgreSQL database with PostGIS. Solara chooses which one to read in this order: the environment variable `TERRA_BR_DSN`, then the database connected in the *Grid store* card. Without either of them there is no store, and every reading that depends on it is refused. The address `postgresql:///terra_br` is what the empty fields of the card stand for, and it applies only after connecting. TERRA reads the same variable and, in its absence, uses this address by default. What the store contains defines what Solara offers.

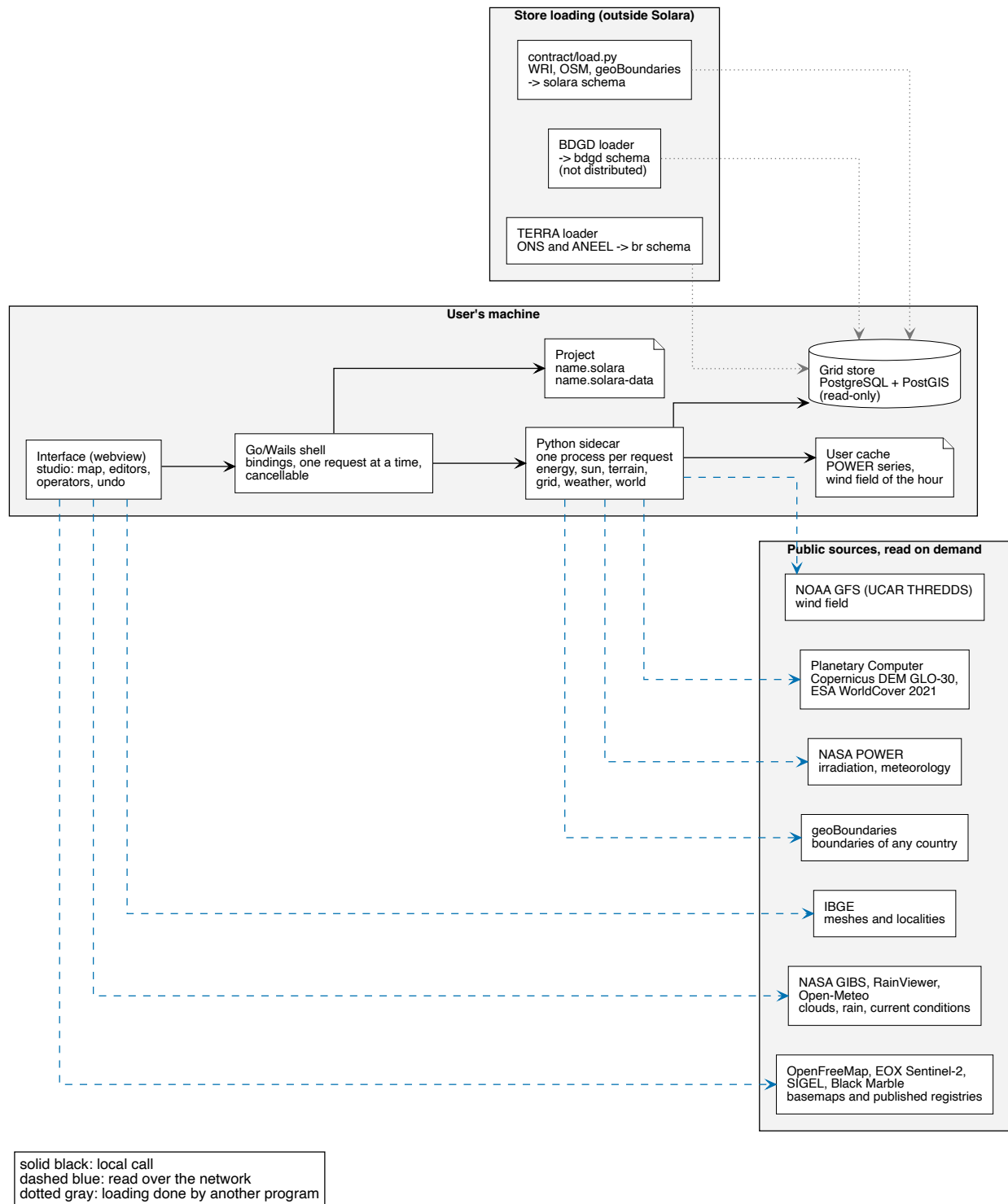


Fig. 6. All the computation and all the state stay on the user's machine; the network supplies only public data. Components surveyed from the Solara code. Solid black line: local call. Dashed blue line: read over the network. Dotted gray line: loading of the store done by another program. BDGD: Base de Dados Geográfica da Distribuidora (geographic database of the distribution utility). GFS: Global Forecast System. WRI: World Resources Institute. OSM: OpenStreetMap.

Table 11

Schemas of the store and what each one enables.

Schema	Content	Enables
br	ANEEL plant register, ONS transmission registry and the ONS registry of curtailment of photovoltaic plants; loaded by TERRA	plant, line and bus layers; grid connection with curtailment
bdgd	consumer units and distributed generators with month-by-month energy, and the sets of each distribution utility, by base year	area consumption; footprint of each registry; consumption by municipality
solara	contract version 1: plants, substations, lines and boundaries of any country	plant and grid layers; grid connection; area catalog

The contract fixes the units (MW, kV, MVA) and the projection (EPSG:4326); the conversion is up to whoever prepares the store. An unknown value is null, not zero. The line declares whether the geometry follows the route of the conductor or is the straight segment between the ends; the distance to a straight segment is a lower bound, and the reading reports this. Curtailment, consumption by distribution utility, concession areas and time series are not in the contract.

7.2 Sources

Table 12

Data sources and terms of use.

Data	Source	Terms
Irradiation and meteorology	NASA POWER	public
Elevation	Copernicus DEM GLO-30, through the Microsoft Planetary Computer	free, with attribution
Permanent water	ESA WorldCover 2021 v200, 10 m, through the Planetary Computer	CC BY 4.0
Plant register	ANEEL (SIGA)	open data
Transmission and curtailment	ONS	open data
Consumption and distributed generation	BDGD, ANEEL	open data
Administrative boundaries	IBGE; geoBoundaries	open data; CC BY 4.0
Clouds	NOAA GOES-East, through NASA GIBS	public
Rain	RainViewer	free, with attribution; requests limited per address
Wind field	NOAA GFS, through UCAR THREDDS	public; research service with no availability guarantee
Current conditions	Open-Meteo	CC BY 4.0; free for non-commercial use
Street basemap	OpenFreeMap, OpenStreetMap	ODbL
Imagery basemap	EOX Sentinel-2 cloudless 2016	CC BY 4.0; from 2017 onwards, CC BY-NC-SA

8 Standing caveats

The caveats in this section apply to the current readings. They reproduce what the sources state, and are not defects to be fixed.

- **Distributed generation in the BDGD is injection, not generation.** The registry carries what passed through the meter of the distribution utility; self-consumption does not pass through it. For Cosern, base year 2024, the median generator reports 51 % of the yield modeled by solar

resource at the same place. The quantity serves a grid question and not a generation question. The definition is not confirmed in the regulation: the requests for the annex of REN 956/2021 and for the BDGD manual received no response.

- **The installed capacity column mixes units.** Some rows are in kW and some in W. The reading treats values above 75 as W, which is a heuristic and does not cover all cases; the reading that reports installed capacity states that it normalized the values.
- **The municipality is the floor of an area.**
- **The footprint of a registry is not a concession area.** It is the union of the sets of the registry and answers where the registry has data, not who holds the concession.
- **The exclusion by height above drainage, in usable ground, is a lower bound.** The elevation model is read over the area and a margin of 2 to 5 km, not over the upstream basin. A channel that enters from outside the margin arrives without its contributing area, and less ground is excluded than would otherwise be. The number appears as “at least”.
- **The water in usable ground is the permanent water mapped in 2021.** Seasonal water, rivers narrower than a few cells, wetlands and whatever changed since then remain subject to the two rules. The sea is counted as ground where the map has no data.
- **Wind screening is raw and not validated.**
- **The connection reading is not a grid access opinion (*parecer de acesso*).**

9 Evolution

The two tables reproduce the Solara work list reported by the owner on 2026-10-05. The list is not in a Solara file, and the column “What it already has” was not checked against the code. A *workstream* is a subject of the list; it can cover part of a product of section 4 or more than one. Solar terrain has no item in the list.

Table 13

Workstreams that already exist and what is missing in each one.

Workstream	Current product	What it already has	What is missing
Ground screening	Usable ground	slope, flooding and permanent water	land cover, mangrove and wetland, protected areas, largest contiguous usable block
Solar resource	Solar resource	irradiation at a point from reanalysis	satellite source, ready-made maps
Photovoltaic generation	Solar resource	reference array of 1 kWp, losses as a single ratio	module and inverter library, detailed losses, horizon at the point
Wind resource	Wind screening	screening at a point with reanalysis at 10 m and 50 m	a better source at hub height, mapped relief and roughness, import of mast measurements
Wind generation	Wind screening	one fixed turbine, raw values	turbine catalog, turbine layout, wake, long-term correction
Grid	Grid connection	lines, substations, point of connection, headroom and curtailment, Brazil only	connection queue, power flow, grids of other countries
Consumption	Area consumption	monthly consumption of an area from the BDGD	hourly load curve of a customer, tariff values

Table 14

Workstreams that do not exist and what each one depends on. “Chaining” is the use of the result of one product as an input of another.

Workstream	Depends on
Financial	photovoltaic or wind generation, and chaining between results
Storage and hybrids	generation, price series and chaining
Market and revenue	price source per country
Environmental impact (wind)	turbine layout
Land and permitting	land tenure base per country
System planning	grid and consumption; it is close to a separate program

10 Open questions

1. **Target customer.** Solara serves those who study sites and areas for generation. The distribution-utility track of the feature catalog depends on private data and on a continuous service, which are outside the stated scope. It remains to be decided whether this track is dropped, postponed or treated as a separate product.
2. **Forecasting models.** No Solara product reads a forecasting model from the experiments of the project. It remains to be defined whether any of them becomes a product and at which scale of place.
3. **BDGD loader.** Area consumption depends on the bdgd schema, and the program that loads it is not part of Solara. Until it is, the product is reproducible only on a machine that already has the store.
4. **Commercial use.** The current conditions reading uses the free tier of Open-Meteo, restricted to non-commercial use, and the imagery basemap is restricted to the 2016 edition, because the editions from 2017 onwards carry a non-commercial license.
5. **Area below the municipality.** No source subdivides a municipality; neighborhoods and census tracts are not reading areas.